

Sales Performance Analysis

Company: Lumen Living (fictional) · Period: January–December 2024 · Data: Fully simulated

Disclaimer: Lumen Living, its products, customers and every figure in this document are simulated for a portfolio project. No statement represents a real business and no causal claims are made.

1. Project Overview

Lumen Living is a simulated DTC e-commerce brand selling five categories across six acquisition channels. Management asked: “Is our growth healthy, and where should we focus next?” This analysis uses 72 monthly-channel rows and focuses on decision support.

2. Executive Summary

Observed facts: revenue grew from \$210,000 in January to \$385,000 in December (+83%), with a November peak of \$412,000. Total revenue was \$3,395,000 across 41,649 orders and 83,823 units; AOV was \$81.52. Organic Search and Paid Search together generated more than half of revenue. Email had the highest listed conversion rate (4.2%) and a \$95 AOV. Paid Social had the lowest listed conversion (1.4%) and AOV (\$65). Home Decor and Kitchen represented 53% of revenue. Repeat rate was 31.3%.

3. Business Problem

The analysis addresses how sales performance changes over time, which channels contribute most, which appear more efficient, which categories drive revenue, where opportunities exist, and which KPIs should be monitored.

4. Analytical Approach

Load the simulated CSV into Excel and SQL; build monthly, channel and category roll-ups; compare revenue, AOV, conversion and repeat rate; stress-test facts versus interpretation; then frame recommendations. AI was used to accelerate exploration and critique while calculations were independently validated.

5. Monthly Performance

Revenue growth is volume-led: January-to-December revenue rose 83% while AOV remained broadly stable around \$81. November was the strongest month and February the notable dip. These observations describe the supplied simulated dataset, not a real market pattern.

6. Channel Analysis

Organic Search is the largest revenue channel. Email combines the highest conversion rate with a high AOV and repeat profile. Paid Social has the lowest listed conversion and AOV while its share grows across the year. These patterns identify areas for investigation; they do not prove causal effects.

7. Product Category Analysis

Home Decor and Kitchen together contribute 53% of revenue. Home Decor and Kitchen peak in November, Outdoor shows a seasonal May–July shape, and Bedding/Lighting are comparatively stable in the supplied dataset.

8. Recommendations

1) Investigate Paid Social efficiency using CAC and ROAS data. 2) Test controlled lifecycle/email expansion. 3) Protect Q4 Home Decor and Kitchen inventory and campaign planning. 4) Explore referral growth. 5) Monitor category concentration. 6) Establish a recurring KPI review.

9. AI-Assisted Workflow

DATA → AI-assisted exploration → pattern identification → human validation → business interpretation → recommendation → decision support. AI helped with pattern detection, structuring, critique and drafting. Final calculations

were recomputed independently in Excel and SQL.

10. BA Skills Demonstrated

Excel modelling · SQL querying · KPI design · funnel thinking · business framing · written case study · AI-assisted workflow · documentation · critical thinking about causality.

11. Limitations

The data is simulated and aggregated at monthly × channel level. There is no customer-level or order-level data, no cost data for CAC/ROAS/margin/LTV, and no causal inference. Associations are observations, not proof of cause.

12. Next Steps

Add spend data for CAC/ROAS; add customer IDs for cohort retention; add order-level data for SKU/margin analysis; test budget or lifecycle interventions; extend the dataset beyond 12 months for stronger seasonality analysis.